

Thrashing is a phenomenon in virtual memory scheme when the processor spends most of its time in swapping pages, rather than executing instructions.

18) What are the four necessary and sufficient conditions behind the deadlock?

These are the 4 conditions:

- 1) Mutual Exclusion Condition: It specifies that the resources involved are non-sharable.
 - 2) Hold and Wait Condition: It specifies that there must be a process that is holding a resource already allocated to it while waiting for additional resource that are currently being held by other processes.
 - 3) No-Preemptive Condition: Resources cannot be taken away while they are being used by processes.
 - 4) Circular Wait Condition: It is an explanation of the second condition. It specifies that the processes in the system form a circular list or a chain where each process in the chain is waiting for a resource held by next process in the chain.
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19) What is a thread?

A thread is a basic unit of CPU utilization. It consists of a thread ID, program counter, register set and a stack.

20) What is FCFS?

FCFS stands for First Come, First Served. It is a type of scheduling algorithm. In this scheme, if a process requests the CPU first, it is allocated to the CPU first. Its implementation is managed by a FIFO queue.

21) What is SMP?

SMP stands for Symmetric MultiProcessing. It is the most common type of multiple processor system. In SMP, each processor runs an identical copy of the